

IoT based Soil Nutrient Monitoring and Analysis System

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Abstract— Traditional soil nutrient monitoring methods are time-consuming and inefficient, relying on manual sampling and delayed laboratory testing, which hinders timely interventions and leads to resource wastage in agricultural practices. This paper presents an IoT-based soil nutrient monitoring and analysis system designed to address these inefficiencies by enabling real-time data collection and analysis. The proposed system utilizes a network of advanced soil sensors that measure key parameters such as nitrogen, phosphorus, potassium (NPK), moisture, and pH levels. The data is wirelessly transmitted to a cloud platform for continuous processing and analysis, providing real-time insights into soil health. The innovation of this system lies in the integration of IoT sensors, cloud computing, and data analytics, which enables precise and timely decision-making for optimal soil management. Compared to recent soil monitoring systems that either lack real-time capabilities or require costly infrastructure, this IoT-based solution offers a more affordable, scalable, and efficient approach. By offering actionable insights, this system significantly improves crop management, reduces overuse of fertilizers, and enhances resource efficiency, ultimately leading to higher yields and more sustainable agricultural practices.

Keywords—Internet of Things(IOT),Soil Nutrients lacking and abundance levels, Smart farming, Soil fertility.

I. INTRODUCTION

The global agricultural sector faces significant challenges related to food security, climate change, and the sustainable use of resources. Soil health is a key factor in crop productivity, yet traditional methods of soil nutrient analysis are often inefficient, time-consuming, and provide only periodic snapshots of soil conditions. With the increasing demands for food production, there is an urgent

need for more precise, real-time monitoring solutions to optimize soil management and improve crop yields. The advent of the **Internet of Things (IoT)** offers a promising solution to these challenges by enabling continuous, real-time data collection from the field, offering insights into soil nutrient levels, moisture, and other essential parameters. Despite the advancements in precision agriculture, current IoT applications in soil nutrient management remain limited, either focusing on specific parameters or lacking integration with comprehensive data analytics systems that can deliver actionable insights to farmers [3].

The farming industry has recently benefited greatly from the growth of IOT technologies, especially in terms of communication infrastructure. In order to correct nutrient deficiencies, farmers are unable to personally monitor and inspect every plant. IOT technology is still helpful, though, and it has helped farmers enter a new phase of smart agriculture[4]. With increased accuracy, sensors are continuously observing crops to identify any unfavourable soil conditions.

Smart agriculture and monitoring play a crucial role in addressing various issues related to crop production, especially in monitoring changes in soil characteristics like soil temperature, soil PH, soil moisture and soil nutrients like nitrogen, phosphorous and potassium and it compares the measured values with the defined nutrients values of a particular crop for high yield[5]. And stores this data in the cloud so that farmer can access these data from anywhere in mobile phone or computer. This helps enhance spatial management practices, leading to increased crop production while minimizing the excessive use of fertilizers[5]. This also leads to sustainable agriculture, which is important for protecting biodiversity, maintaining natural resources, and cutting greenhouse gas emissions.

II. LITERATURE SURVEY

Based on various kinds of crop and field types, researchers have proposed a variety of approaches,

structures, and equipment to monitor and relay crop information at different growth phases.

Raneesha Madhushanki, A.A. The applications of Internet of Things (IoT) for enhancing agricultural productivity is the topic of this study. IoT gathers and analyzes sensor data from the fields to boost productivity and minimize the need for human labor. Most significant IoT subvertical identified to be water management, which was followed by crop management, smart farming, animal management, and irrigation management at the same intensity. The environmental temperature, environmental humidity, soil moisture, and soil pH had the most influence on sensor data. Wi-Fi is thought to be the most widely deployed IoT technology in agriculture[6].

"Internet of Things (IoT) in Agriculture Applications, Challenges, and Future Directions" by N. Misra. This comprehensive review discusses the applications of IoT in agriculture, including precision farming, smart irrigation, and crop monitoring. It explores the challenges faced in implementing IoT solutions in agriculture and proposes future research directions [7].

Recent Advances in Internet of Things (IoT) technologies for Agriculture and Precision farming by S.Mukherjee et.al presents recent advances in IoT technologies for agriculture and precision farming. It covers topics such as wireless sensor networks, cloud computing, and data analytics, highlighting their role in improving agricultural productivity and sustainability[9].

Precision Agriculture Techniques and Practices: A Review by R. Kaur [10] et.al provides an overview of precision agriculture techniques and practices, including GPS, GIS, remote sensing, and IoT. It discusses the role of precision agriculture in improving resource efficiency, crop yield, and environmental sustainability.

"Wireless Sensor Networks for Agriculture: A Review" by S. Singh et.al presents paper discusses the use of wireless sensor networks (WSNs) in agriculture for monitoring soil conditions, crop growth, and environmental parameters. It presents case studies of WSN- based agricultural applications and discusses the challenges and opportunities in this field[11].

Internet of Things (IoT)-Based Smart Agriculture: Toward Making the Fields Talk by M. K. Patel et.al paper provides an overview of IoT-based smart agriculture systems, focusing on sensor technologies, communication protocols, and data analytics. It discusses the benefits of IoT in agriculture and presents case studies of IoT- based agricultural solutions[12].

Traditional soil testing methods are often time-consuming, costly, and prone to inconsistencies due to spatial and temporal variability in soil properties. As a result, farmers and agricultural experts face challenges in maintaining optimal soil health, which directly impacts crop yield and sustainability. There is a pressing need for a more efficient, real-time, and cost-effective solution to monitor soil nutrient levels and provide actionable insights. This research proposes the development of an **IoT-Based Soil Nutrients and Analysis System**, aimed at providing real-time, remote monitoring and analysis of key soil nutrients

(e.g., nitrogen, phosphorus, potassium). By leveraging IoT sensors, cloud computing, and data analytics, the system will enable continuous soil health monitoring, optimize fertilizer usage, and improve crop yield predictions. The research will explore the design and implementation of an integrated IoT sensor network, the real-time transmission of soil data, and the development of a decision-support system for farmers. The objectives of this research are:

1. To design and develop a low-cost, scalable IoT system capable of accurately measuring soil nutrients in real-time.
2. To analyze the effectiveness of IoT-based data collection compared to traditional soil sampling methods.
3. To create a data-driven decision-making platform that provides farmers with real-time feedback and recommendations on nutrient management.
4. To assess the impact of real-time soil monitoring on crop yield, resource efficiency, and sustainable farming practices.

By introducing a smart soil nutrient analysis system, this research will contribute to the ongoing digital transformation of agriculture. The outcomes of this study are expected to reduce input costs for farmers, minimize environmental impact due to excessive fertilizer use, and promote more sustainable and productive farming practices. The IoT-based approach will also enhance precision agriculture by providing data that can lead to better decision-making in nutrient management and soil health maintenance.

III. METHODOLOGY

In the proposed system we were making use of Internet of Things (IoT) based technology. Here we use Arduino Uno Rev3 ,ESP32S and thingsboard cloud platform are the main functioning units.

The Arduino uno receives the data from the all soil parameters and nutrients measuring sensors like NPK sensor, soil temperature sensor ,soil moisture sensor, PH sensor. The received data from the sensors to transmitted to the ESP32S processor which contains the wifi module .so that the Arduino uno acts as a sender whereas the ESP32S acts as a receiver. The data received by the ESP32S is stored in a IOT platform Thingsboard cloud which can be accessible by the farmer in mobile phones or computers. The soil parameters measurement is continuously done and these values are stored in the cloud. These measured values are being analysed in the Arduino code which has the predefined values of different crops.

Through seamless integration with IoT platforms such as ThingsBoard, farmers gain access to valuable insights into soil conditions, allowing for informed decision

regarding irrigation scheduling, fertilization practices, pH adjustment, and other agronomic interventions. The visual feedback provided by the LED strip enhances on-site monitoring, enabling farmers to quickly assess soil health and take timely actions to optimize crop yields while promoting sustainable land management practices

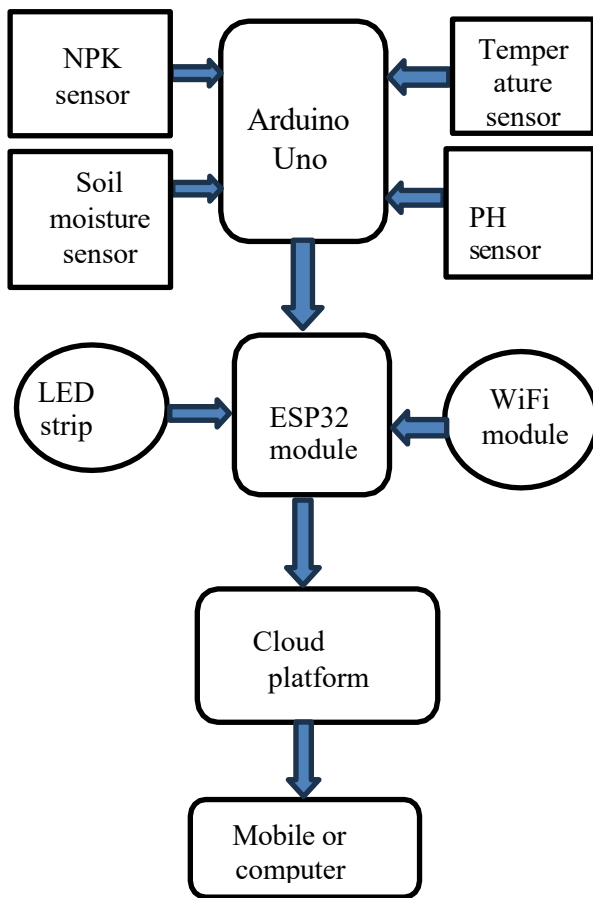


Fig. 1. Block Diagram of system

IV. PROPOSED METHOD

A. OVERVIEW

IoT enables real-time monitoring and data analysis by integrating sensors with cloud platforms. It allows for continuous data collection from the soil, which is analyzed and transmitted wirelessly to provide insights into nutrient levels, moisture, and temperature. This data helps optimize farming practices

The integrated soil monitoring system operates through a series of interconnected components, seamlessly capturing, processing, and transmitting soil-related data for real-time monitoring and analysis. Below is an overview of the working principles:

1. **Sensor data collection:** - The system comprises multiple soil sensors, including NPK (Nitrogen, Phosphorus, Potassium), temperature (DSB180), soil moisture, and pH sensors. These sensors are strategically placed in the soil to capture relevant parameter vital for crop growth and soil health. Each sensor continuously measures its respective parameter and provides analog or digital output signal proportional to the measured value.
2. **Arduino data processing:** - The analog and

digital output signals from the sensors are interfaced with an Arduino microcontroller board. The Arduino board reads these signals and converts them into meaning data values using data values using appropriate algorithms or calibration curve. The processed sensor data is then stored in variables within the Arduino's memory for further processing and transmission.

3. **Communication between Arduino and ESP32S:-** The Arduino board communicates with an ESP32 module via the I2C (Inter-Integrated Circuit) protocol. The I2C protocol allows for bi-directional communication between the Arduino and ESP32, enabling seamless data exchange. Through I2C, the Arduino sends the processed sensor data to the ESP32 for further processing and transmission
4. **Wireless transmission to IOT platform:-** The ESP32 module serves as a bridge between the Arduino and the Internet, facilitating IoT connectivity. It establishes a Wi-Fi connection with the local network and connects to the designated IoT platform, such as ThingsBoard. The ESP32 securely transmits the sensor data received from the Arduino to the IoT platform using communication protocols like MQTT or HTTP.
5. **Integration with IOT platform:-** Upon receiving the sensor data, the IoT platform, such as ThingsBoard, stores the data securely in its database. The platform provides a user-friendly dashboard interface for real-time monitoring and analysis of soil parameters. Farmers can remotely access the data via web or mobile applications, empowering people to decide on fertilization, irrigation, and other farming operations with knowledge.
6. **Implementing the Soil Nutrient Analysis System:** The system can be implemented by placing IoT-enabled sensors in the soil at various locations across the field. These sensors will send data to a central server via an ESP module. The data is then analyzed using cloud computing platforms, and the results are displayed on a dashboard or app for the farmer to make decisions.
7. **Enhancing the Monitoring Process:** The monitoring process can be enhanced by incorporating more advanced sensors that measure additional soil parameters (e.g., micronutrient levels), using solar power for sustainability, and integrating predictive analytics to suggest crop intervention

By leveraging Arduino and ESP32 platforms, along with sensor technology and IoT connectivity, the integrated soil monitoring system offers a comprehensive solution for precision agriculture, empowering farmers with actionable insights to optimize crop production and sustainable land management practices. While selecting the appropriate technique for the research several critical issues like Soil type and environmental conditions, Energy efficiency and power supply, Cost ,Scalability, Sensor Accuracy and precision, must be considered to ensure the chosen approach is effective, efficient, and aligned with the project's goals.



Fig2. Overview of IOT system.

B. HARDWARE COMPONENTS USED

TABLE1.HARDWARE USED

Components	Specifications
Arduino uno board	Operating range-5V
ESP32S module	Memory 320 KB SRAM
RS485 communication module	32 unit loads
LED strip	6 LEDS
Soil moisture sensor	
Temperature sensor(DSB1820)	Measuring range:- -55C to85C

Arduino Uno Board:- Arduino Uno Atmega328p is used. Its operating voltage is 5V. The selection of Arduino is based on its cost-effectiveness, ease of integration with different sensors, and support for wireless modules such as ESP for Internet of Things applications. It is a flexible platform that works well for IoT solution deployment and experimentation in agriculture.

ESP32S Module:- The role of ESP32S module in this system, it acts as a The wireless communication module in this system is an ESP (such as an ESP8266 or an ESP32). It allows sensor data to be wirelessly transmitted to databases or cloud servers. Because ESP modules are inexpensive, lightweight, and power-efficient, they are perfect for Internet of Things applications in agriculture

NPK sensor:- This sensor measures the levels of essential nutrients - Nitrogen (N), Phosphorus (P), and Potassium(K) in the soil, which are vital for plant growth and development.

TABLE2.NPK SENSOR SPECIFICATIONS

Measuring range	0-1999mg/kg
Supply voltage	5-24V DC
Working range	0-50 C
Resolution	1mg/kg



Fig. 3. NPK Sensor

PH sensor:- The pH sensor measures the acidity or alkalinity of the soil, providing insights into soil fertility and nutrient availability. Maintaining the appropriate pH level is crucial for optimal nutrient uptake by plants and microbial activity in the soil.



Fig. 4. PH sensor

V. RESULT AND DISCUSSION

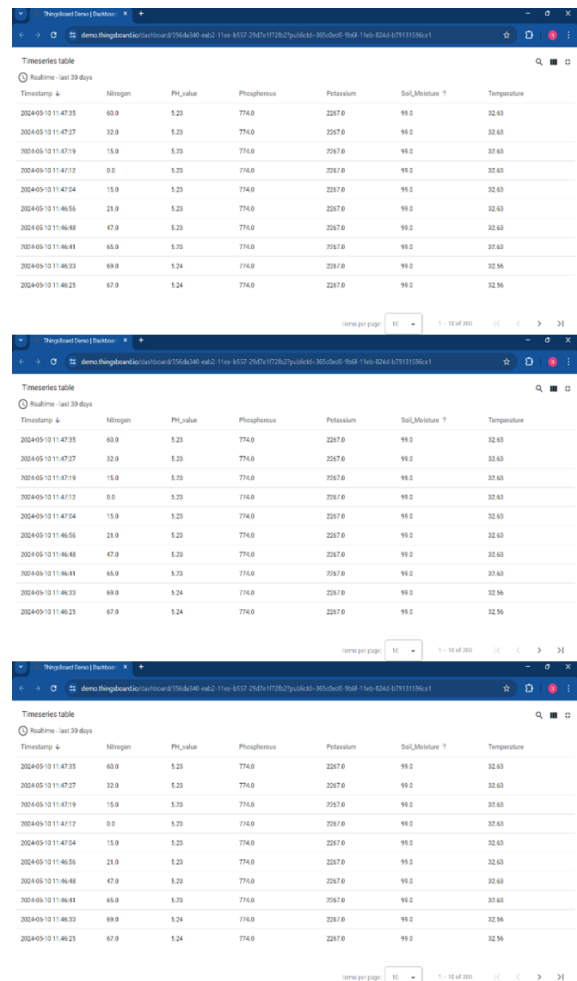


Fig 5. Result of measured values of soil parameter

Fig2 illustrate the recorded data of soil characteristics such as Nitrogen, phosphorous, potassium, PH value, Soil moisture and soil temperature. The system consistently monitors these parameters and saves them in a cloud along with date and time. This data is further being analysed in Arduino software with python code. The input data for the python code in the Arduino software is the monitored data in Fig2. This code includes the preset values of soil parameters in order to optimize the yield of various food crops such as paddy and wheat. By comparing the measured values with the specified values for the desired crop, we can determine any discrepancies which indicates an excess or deficiency in the soil parameters. Furthermore, this analysis is conducted continuously and the results are stored in the cloud.

Improving Accuracy :- Reducing interference in wireless communication, utilizing error-correction algorithms, and calibrating high-precision sensors on a regular basis are all ways to attain high accuracy. Accuracy can also be increased by utilizing stronger data processing methods, such as AI.

Improving Precision:- Precision can be improved by using high-quality, calibrated sensors, advanced machine learning algorithms, and AI-based data analytics to predict nutrient deficiencies and optimize fertilizer application. Moreover, leveraging GPS-based location services can provide precise data for specific regions within a field.

Optimizing Crop Yields: Crop yields can be optimized by providing the right amount of nutrients based on real-time data and predictive insights. IoT systems can help farmers apply fertilizers and water more efficiently, reduce wastage, and ensure that the crops receive the right inputs at the right time

VI. COMPARISON

Table 3. Comparative study between traditional methods and IOT based system[12][13].

Parameter	Traditional Methods	IoT-Based Soil Nutrients Monitoring System
Data Collection Frequency	Periodic (seasonal or specific intervals)	Continuous, real-time monitoring 24/7
Real-Time Monitoring	Not available (lab analysis takes days/weeks)	Real-time data availability for immediate analysis
Accuracy and Precision	High in lab settings, but may miss	Moderate to high (depends on sensor

	spatial variability	type, covers spatial variability)
Spatial Coverage	Low (relies on manual sampling from limited areas)	High (multiple sensors can cover different areas of the farm)
Cost	High (due to repeated lab tests and labor costs)	High initial investment but lower long-term costs (minimal human intervention needed)
Environmental Impact	High (due to delayed feedback, overuse of fertilizers)	Low (real-time adjustments reduce overuse of fertilizers and minimize runoff)
Ease of Use	Labor-intensive, requires expert knowledge	Easy to use, automated data collection with user-friendly interfaces (e.g., mobile apps)
Decision-Making Impact	Delayed (based on lab reports that may be outdated)	High (decisions can be made in real time based on continuous monitoring)
Scalability	Low (requires significant labor and costs for large-scale farms)	High (can be scaled easily with additional sensors)
Maintenance Requirements	Requires repeated sampling and testing	Low maintenance once installed, occasional calibration required

Adaptability to Local Conditions	Low (does not account for local variations in real-time)	High (sensors can be placed in specific zones to adapt to soil variability)
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VII. CONCLUSION

In terms of paper presented, the integrated soil monitoring system developed in this project represents a significant advancement in precision agriculture, offering farmers a comprehensive solution for real-time monitoring and management of soil health. By leveraging Arduino and ESP32 platforms, along with a range of sensors including NPK, temperature, soil moisture, and pH sensors, the system enables precise measurement and analysis of key soil parameters crucial for crop growth and productivity. Overall, the integrated soil monitoring system empowers farmers with actionable data, facilitating precision agriculture and driving efficiency, productivity, and sustainability in modern farming practices. After monitoring and analysing the soil parameters we can identify any excesses or deficiencies in specific nutrients values. This allows for the possibility of automatically supplying the necessary amounts of nutrients rather than relying on manual intervention.

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